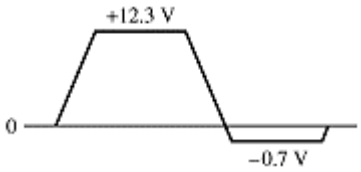
Model answer of sheet 6

1	$\Delta I_Z = 50\text{ mA} - 25\text{ mA} = 25\text{ mA}$ $\Delta V_Z = \Delta I_Z Z_Z = (+25\text{ mA})(15\ \Omega) = +0.375\text{ V}$ $V_Z = V_Z + \Delta V_Z = 4.7\text{ V} + 0.375\text{ V} = \mathbf{5.08\text{ V}}$
2	$V_{IN(min)} = V_Z + I_{ZK}R = 14\text{ V} + (1.5\text{ mA})(560\ \Omega) = \mathbf{14.8\text{ V}}$
3	$\Delta V_Z = (I_Z - I_{ZK})Z_Z = (28.5\text{ mA})(20\ \Omega) = 0.57\text{ V}$ $V_{OUT} = V_Z - \Delta V_Z = 14\text{ V} - 0.57\text{ V} = 13.43\text{ V}$ $V_{IN(min)} = I_{ZK}R + V_{OUT} = (1.5\text{ mA})(560\ \Omega) + 13.43\text{ V} = \mathbf{14.3\text{ V}}$
4	$\Delta V_Z = I_Z Z_Z = (40\text{ mA} - 30\text{ mA})(30\ \Omega) = 0.3\text{ V}$ $V_Z = 12\text{ V} + \Delta V_Z = 12\text{ V} + 0.3\text{ V} = 12.3\text{ V}$ $R = \dfrac{V_{IN} - V_Z}{40\text{ mA}} = \dfrac{18\text{ V} - 12.3\text{ V}}{40\text{ mA}} = \mathbf{143\ \Omega}$
5	$V_Z \cong 12\text{ V} + 0.3\text{ V} = 12.3\text{ V}$ 
6	$V_{Z(min)} = V_Z - \Delta I_Z Z_Z = 5.1\text{ V} - (49\text{ mA} - 1\text{ mA})(7\ \Omega)$ $= 5.1\text{ V} - (48\text{ mA})(7\ \Omega) = 5.1\text{ V} - 0.336\text{ V} = 4.76\text{ V}$ $V_R = 8\text{ V} - 4.76\text{ V} = 3.24\text{ V}$ $I_T = \dfrac{V_R}{R} = \dfrac{3.24\text{ V}}{22\ \Omega} = 147\text{ mA}$ $I_{L(max)} = 147\text{ mA} - 1\text{ mA} = \mathbf{146\text{ mA}}$ $V_{Z(max)} = 5.1\text{ V} + (70\text{ mA} - 49\text{ mA})(7\ \Omega) = 5.1\text{ V} + 147\text{ mV} = 5.25\text{ V}$ $V_R = 8\text{ V} - 5.25\text{ V} = 2.75\text{ V}$ $I_T = \dfrac{2.75\text{ V}}{22\ \Omega} = 125\text{ mA}$ $I_{L(min)} = 125\text{ mA} - 70\text{ mA} = \mathbf{55\text{ mA}}$
7	% Load regulation $= \dfrac{V_{Z(max)} - V_{Z(min)}}{V_{Z(min)}} \times 100\% = \dfrac{5.25\text{ V} - 4.76\text{ V}}{4.76\text{ V}} \times 100\% = \mathbf{10.3\%}$
8	$\boldsymbol{V_{RL} = I_{Lmax} \cdot R_{min}}$ $\boldsymbol{or,\ \ R_{min} = \dfrac{V_{RL}}{I_{Lmax}} = \dfrac{5}{I_{Lmax}} \quad ------(1)}$ $\boldsymbol{I = I_{zk} + I_{Lmax}}$ $\boldsymbol{or,\quad I_{Lmax} = I - I_{zk} = 50 - 10 = 40mA}$ $\boldsymbol{R_{min} = \dfrac{5}{40 \times 10^{-3}} = \dfrac{5000}{40} = \dfrac{500}{4} = 125\Omega}$